

Best Practices

Introduction to Programming

StayAhead Training; Sep, 2024

Program Design

Best Practices

- Low-level design = algorithms <!-- this for now!
- High-level design = architecture
- **Algorithm:** a sequence of instructions for solving a problem

Algorithm Design

Best Practices

1. State the problem
 - with assumptions
 - in plain English
 - and in detail
2. Devise a solution
 - start with the inputs and outputs
 - solve the problem manually
 - look for patterns e.g. conditions, repetition
 - write pseudocode

Coding Conventions

Best Practices

- Whitespace between tokens
- Indenting
- Line length
- Naming
- Commenting
- Curly brace placement
- Quotation marks

Coding Style

Best Practices

- Consistent
- Readable
- Testable
- Maintainable (DRY, KISS, YAGNI, and SOC)
- Flexible
- Scalable

Unit Testing

Best Practices

- Unit: a module/class/function
- Unit test: some code to verify the behaviour of a unit
- Robustness and regression
- TDD: test driven development (code the tests first)
- "If it ain't tested, it's broken."